ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

POWER MACHINES N6

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central N6 Power Machines N6 paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct methods and clearly labelled diagrams.

QUESTION 1

1.1.1 $Q = m(h - h_f) = 5\,000(2\,800 - 335)$ ✓✓ $= 12\,325\,000$ kJ/h ✓ (3)

1.1.2 Heat supplied $= 650 \times 28\,000 = 18\,200\,000$ kJ/h ✓; $\eta = \frac{12\,325\,000}{18\,200\,000} \times 100$ ✓ $= 67{,}72\%$ ✓ (3)

1.1.3 Equivalent evaporation $= \frac{12\,325\,000}{2\,257}$ ✓✓ $= 5\,460{,}79$ kg/h ✓ (3)

1.1.4 $\frac{5\,460{,}79}{650}$ ✓✓ $= 8{,}40$ kg steam per kg coal ✓ (3)

1.2 (a) Economiser – uses the heat of the flue gases to preheat the feedwater ✓✓ (b) superheater – heats the saturated steam above its saturation temperature (dry, superheated steam) ✓✓ (c) air preheater – uses flue gases to heat the combustion air, improving combustion and efficiency ✓✓ (6)

1.3 Any TWO: safety valve ✓; water level gauge (gauge glass) ✓; pressure gauge; low-water alarm; fusible plug (2)

[20]

QUESTION 2

2.1.1 Boiler → turbine → condenser → feed pump → back to the boiler, with arrows and labels ✓✓; heat input at the boiler, work output at the turbine, heat rejected at the condenser ✓ (3)

2.1.2 $W_T = 3\,230 - 2\,210 = 1\,020$ kJ/kg ✓✓; $W_{net} = 1\,020 - 4 = 1\,016$ kJ/kg ✓ (3)

2.1.3 $Q_{in} = 3\,230 - (192 + 4)$ ✓ $= 3\,034$ kJ/kg ✓ (2)

2.1.4 $\eta = \frac{1\,016}{3\,034} \times 100$ ✓ $= 33{,}49\%$ ✓ (2)

2.1.5 $m = \frac{30\,000}{1\,016}$ ✓ $= 29{,}53$ kg/s ✓; SSC $= \frac{3\,600}{1\,016}$ ✓ $= 3{,}54$ kg/kWh ✓ (4)

2.2 Impulse turbine: the steam expands (pressure drops) only in the fixed nozzles ✓; the moving blades only change the direction of the high-velocity steam, so pressure is constant across the blades ✓. Reaction turbine: the steam expands in both the fixed and the moving blades ✓; the moving blades act as nozzles and are driven by the reaction of the accelerating steam ✓ (4)

2.3 To reduce the very high rotor speed and use the energy efficiently in stages ✓; any ONE: pressure compounding (Rateau), velocity compounding (Curtis), pressure–velocity compounding ✓ (2)

[20]

QUESTION 3

3.1.1 $A = \frac{\pi}{4}(0{,}1)^2 = 7{,}854 \times 10^{-3}\ m^2$ ✓; $n = \frac{3\,000}{2 \times 60} = 25$ power strokes/s per cylinder ✓; $IP = p_m LAnk$ ✓ $= 750 \times 0{,}12 \times 7{,}854 \times 10^{-3} \times 25 \times 4$ ✓ $= 70{,}69$ kW ✓ (5)

3.1.2 $BP = \frac{2\pi NT}{60} = \frac{2\pi \times 3\,000 \times 180}{60}$ ✓✓ $= 56{,}55$ kW ✓ (3)

3.1.3 $\eta_m = \frac{56{,}55}{70{,}69} \times 100$ ✓ $= 80\%$ ✓ (2)

3.1.4 Heat supplied $= \frac{16 \times 44\,000}{3\,600}$ ✓✓ $= 195{,}56$ kW ✓; $\eta_{bt} = \frac{56{,}55}{195{,}56} \times 100 = 28{,}92\%$ ✓ (4)

3.1.5 $bsfc = \frac{16}{56{,}55}$ ✓ $= 0{,}28$ kg/kWh ✓ (2)

3.2 Any TWO (2 marks each): petrol engine uses a spark plug, diesel uses compression ignition ✓✓; petrol has a lower compression ratio (about 8–12) than diesel (about 15–22) ✓✓; petrol is mixed with air before entry (or port injection), diesel fuel is injected directly into hot compressed air; diesel engines are usually more efficient and heavier (4)

[20]

QUESTION 4

4.1.1 $W = \frac{n}{n-1} p_1 V_1 [(\frac{p_2}{p_1})^{(n-1)/n} - 1]$ ✓ $= \frac{1{,}3}{0{,}3} \times 100 \times 2 \times [8^{0{,}2308} - 1]$ ✓✓ $= 533{,}75$ kJ/min ✓ $= 8{,}90$ kW ✓ (5)

4.1.2 $W_{iso} = p_1 V_1 \ln \frac{p_2}{p_1} = 100 \times 2 \times \ln 8$ ✓✓ $= 415{,}89$ kJ/min $= 6{,}93$ kW ✓ (3)

4.1.3 $\eta_{iso} = \frac{6{,}93}{8{,}90} \times 100$ ✓ $= 77{,}92\%$ ✓ (2)

4.1.4 $T_2 = T_1 (\frac{p_2}{p_1})^{(n-1)/n}$ ✓ $= 293 \times 8^{0{,}2308}$ ✓ $= 473{,}45$ K ✓ $= 200{,}45$ °C ✓ (4)

4.2 Any THREE explained (2 marks each): less work is needed because compression is closer to isothermal ✓✓; lower delivery temperature, so better lubrication and safety ✓✓; better volumetric efficiency because the pressure ratio per stage is smaller; moisture can be removed in the intercooler; lighter, better-balanced machine (6)

[20]

QUESTION 5

5.1.1 $RE = 395 - 250$ ✓ $= 145$ kJ/kg ✓ (2)

5.1.2 $W_c = 425 - 395 = 30$ kJ/kg ✓; $COP = \frac{145}{30}$ ✓ $= 4{,}83$ ✓ (3)

5.1.3 $m = \frac{10}{145}$ ✓ $= 0{,}069\ \text{kg/s}$ ✓ (2)

5.1.4 $\text{P} = 0{,}069 \times 30$ ✓ $= 2{,}07\ \text{kW}$ ✓; $\text{Q}_{\text{rej}} = 0{,}069 \times (425 - 250)$ ✓ $= 12{,}07\ \text{kW}$ ✓ (4)

5.2 Diagram: compressor → condenser → expansion (throttle) valve → evaporator → compressor, with flow direction ✓✓; compressor – raises the pressure and temperature of the refrigerant vapour ✓✓; condenser – rejects heat so the vapour condenses to liquid ✓; expansion valve – lowers the pressure and controls the flow of refrigerant into the evaporator ✓✓; evaporator – absorbs heat from the space to be cooled, so the refrigerant evaporates ✓✓ (9)

[20]

TOTAL: 100

END OF MARKING GUIDELINE